identifying faults using AI-driven image analysis and predictive maintenance models.

4. *Agriculture and precision farming*: AI-equipped UAVs monitor soil health, crop growth, and irrigation patterns, applying fertilizers or pesticides precisely where needed, reducing resource waste and improving yield.
5. *Public safety and emergency response*: In disaster or crisis zones, drones assist in search and rescue operations, traffic control, and law enforcement with real-time AI analytics and 3D situational mapping.

6.4.4 Communication and technological applications

Fifth-generation UAVs also play a pivotal role in global communication networks and technological infrastructure. Their integration into AI-powered cloud ecosystems and quantum-encrypted data frameworks has redefined connectivity standards for both defense and civilian systems.

1. *Aerial communication relays*: UAVs function as airborne 6G nodes, establishing temporary communication grids in disaster zones, rural areas, or military operations.
2. *Quantum communication networks*: Drones using quantum key distribution (QKD) create secure communication channels resistant to cyber interception, ensuring data integrity for government and enterprise applications.
3. *Edge computing and cloud integration*: Equipped with onboard AI processors, these drones perform edge analytics, processing data locally and transmitting only essential information to the cloud, reducing latency and bandwidth load.
4. *Digital twin and smart network support*: UAVs integrated with IoT systems provide real-time monitoring and predictive maintenance for